$$Tr\ |\phi\rangle\langle\phi|\mathbf{L} = \langle\phi|\mathbf{L}|\phi\rangle.$$

What if there is a 50 percent probability that she prepared $|\psi\rangle$ and a 50 percent probability that she prepared $|\varphi\rangle$? Obviously, the expectation value is

$$\langle\mathbf{L}\rangle = \frac{1}{2}Tr\ |\psi\rangle\langle\psi|\mathbf{L} + \frac{1}{2}Tr\ |\phi\rangle\langle\phi|\mathbf{L}.$$

All we are doing is averaging over Bob's ignorance of the state prepared by Alice.

But now we can combine the terms into a single expression by defining a density matrix ρ that encodes Bob's knowledge. In this case the density matrix is half the projection operator onto $|\varphi\rangle$ plus half the projection operator onto $|\psi\rangle$,

$$\rho = \frac{1}{2}|\psi\rangle\langle\psi| + \frac{1}{2}|\phi\rangle\langle\phi|.$$

We've now packaged all of Bob's knowledge of the system into a single operator ρ. At this point, the rule to compute expectation values becomes very simple:

$$\langle L\rangle = Tr\ \rho\mathbf{L}. \qquad (7.13)$$

We can generalize this. Suppose that Alice tells Bob that she has prepared one of several states—call them $|\varphi_1\rangle$, $|\varphi_2\rangle$, $|\varphi_3\rangle$, and so on. Moreover, she specifies probabilities P_1, P_2, P_3, … for each of these states. Bob can still package all his knowledge into a density matrix:

$$\rho = P_1|\phi_1\rangle\langle\phi_1| + P_2|\phi_2\rangle\langle\phi_2| + P_3|\phi_3\rangle\langle\phi_3| + \ldots.$$